Don’t forget the attendance check!

Testing (Practice)

Week 7-2

Any fool can write code that a computer can understand. Good programmers write code that humans can understand.

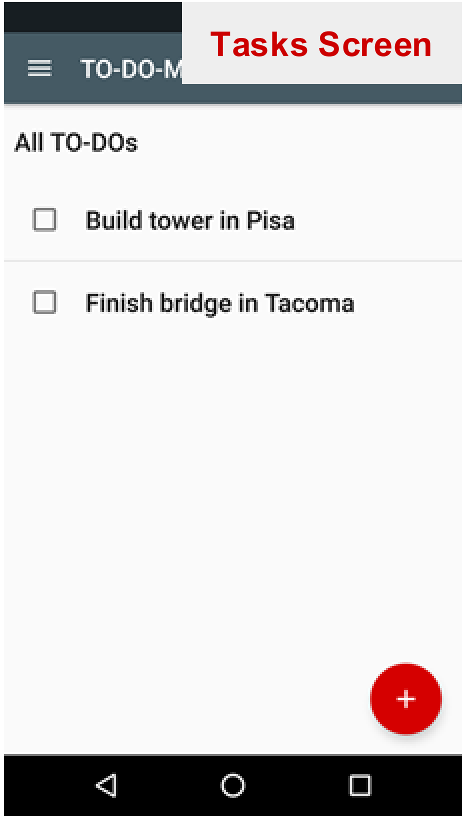
— Martin Fowler

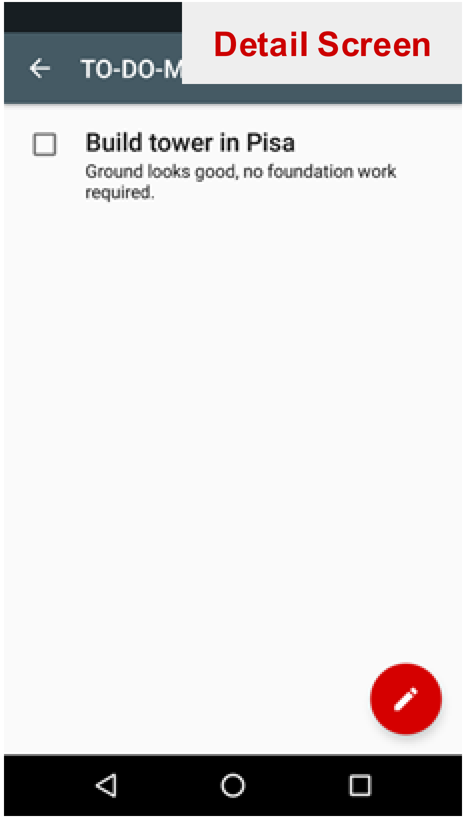
Objectives

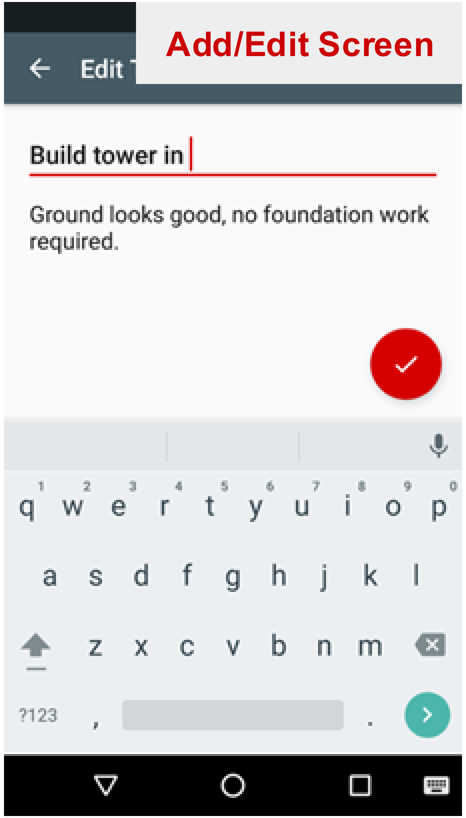
- App Testing Practice using Todo app 😉
- Unit Tests in Two Layers:
  - **Model** Testing (Junit)
  - **ViewModel** Testing (Mockito + Hilt) ***- today's main goal!***

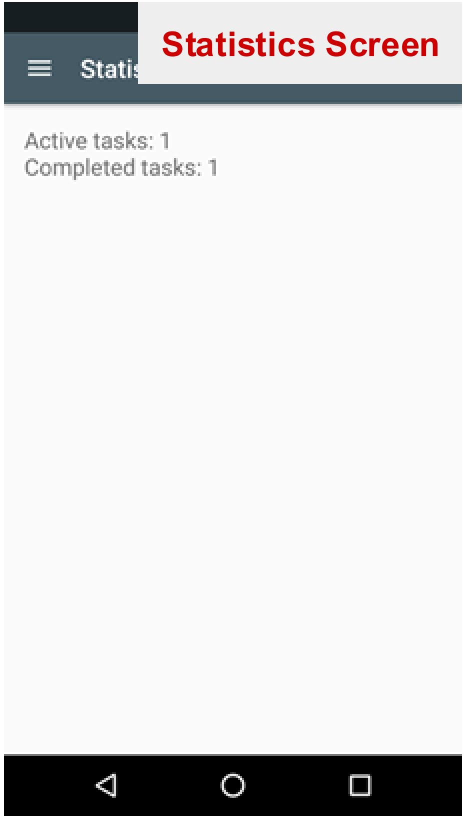
Recap: App Specification

- [App skeleton]
- [App specification]









Recap: Get the Skeleton Code

- Use Git
  - Clone the skeleton repository in the previous slide
- Or you could download the raw zip file

- **Submission Preparation**
  - Create a `submission` directory
  - Make subdirectories **`exercise01 - exercise04`**
  - You have to submit these files at the end of this week!

Recap: Testing Strategy

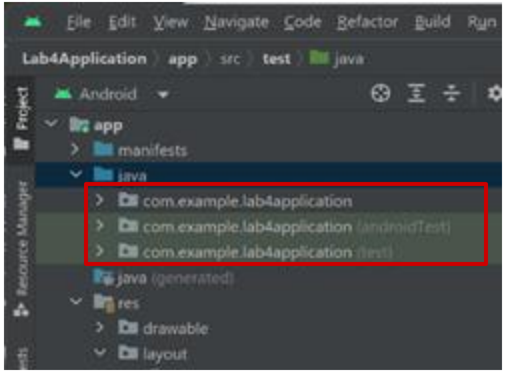
This week's scope!

1. First you'll **unit test** the model.
2. Then you'll use a **test double** in the view model, which is necessary for unit testing and integration testing the view model.
3. Next, you'll learn to write **integration tests** for fragments and their view models.
4. Finally, you'll learn to write **integration tests** that include the Navigation component.

https://developer.android.com/codelabs/advanced-android-kotlin-training-testing-test-doubles#2

Android Studio: Testing Interface

Test Source Sets



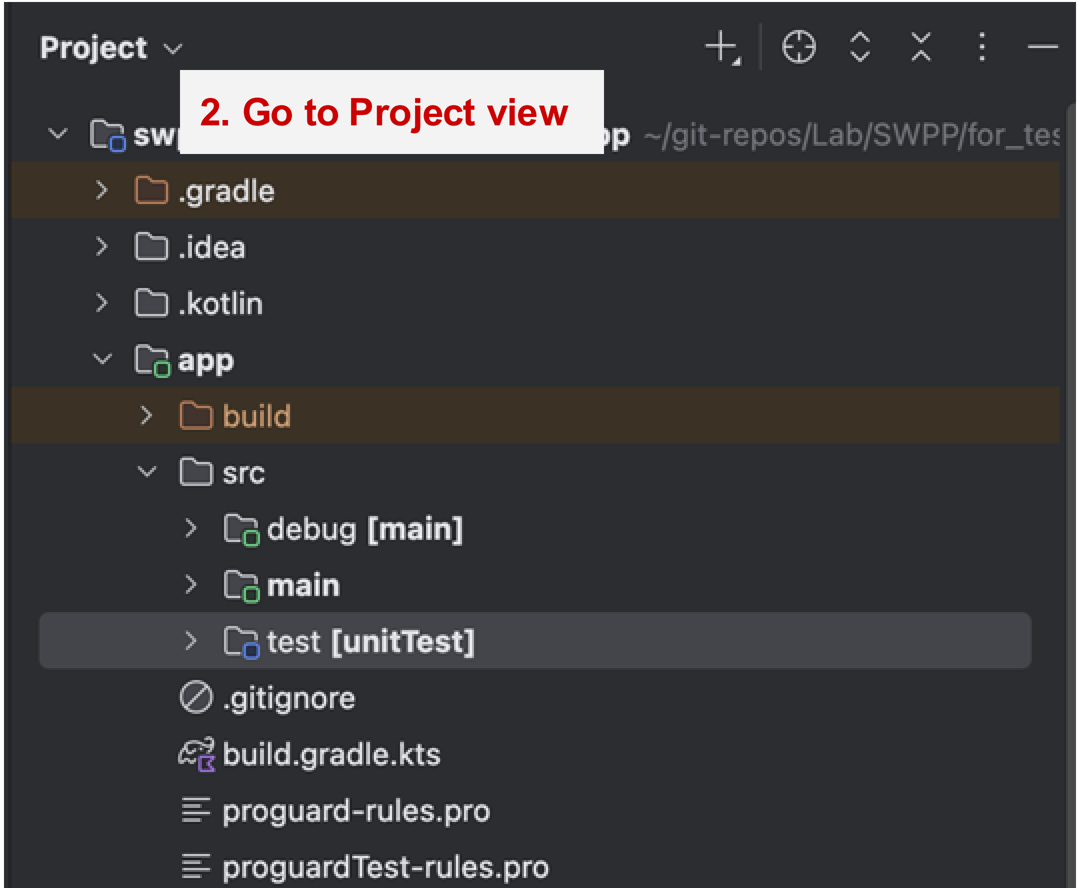
3 **source sets**:

- `main`: contains app code
- `androidTest`: contains instrumented tests
  - Runs on real/emulated device
  - Slow
  - High fidelity
  - For integration, E2E, UI tests
- **`test`**: contains local tests
  - Runs on JVM (not on real/emulated device)
  - Fast
  - Low fidelity
  - For unit tests

Tip: Finding the Test Directories (1/2)

1. First build!

2. Go to Project view



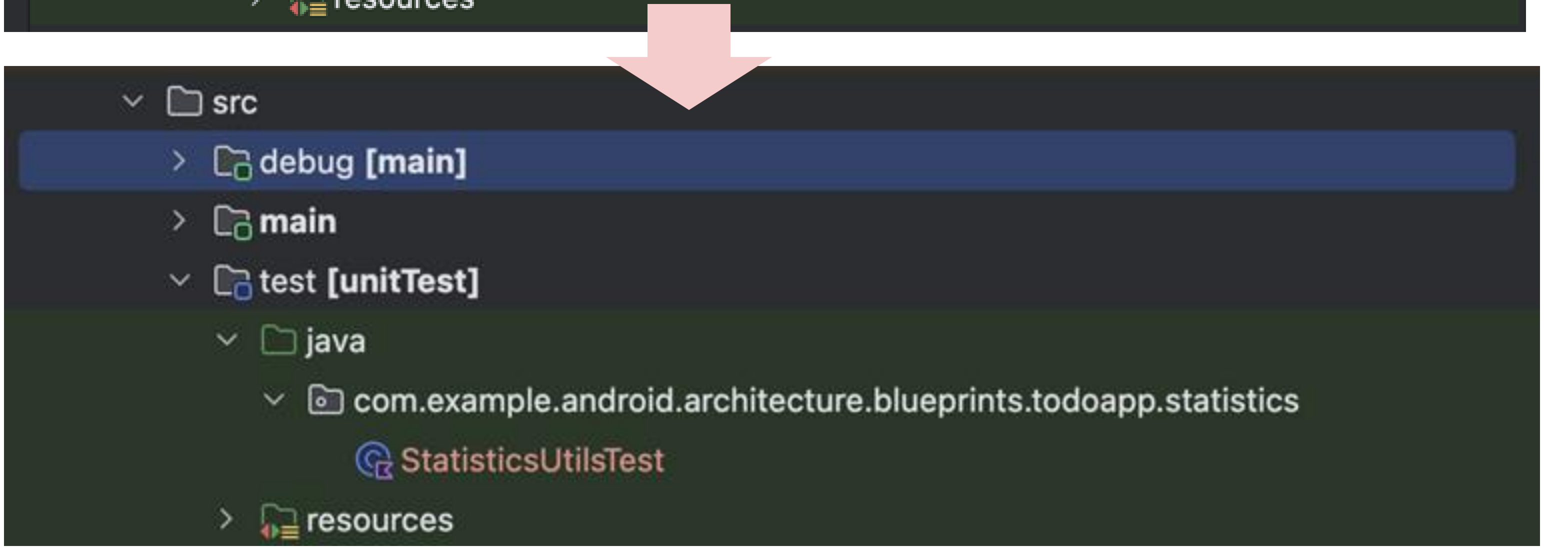
3. Check `app/src/test`

If you don’t have one, create `app/src/test/java/` manually

Tip: Finding the Test Directories (2/2)

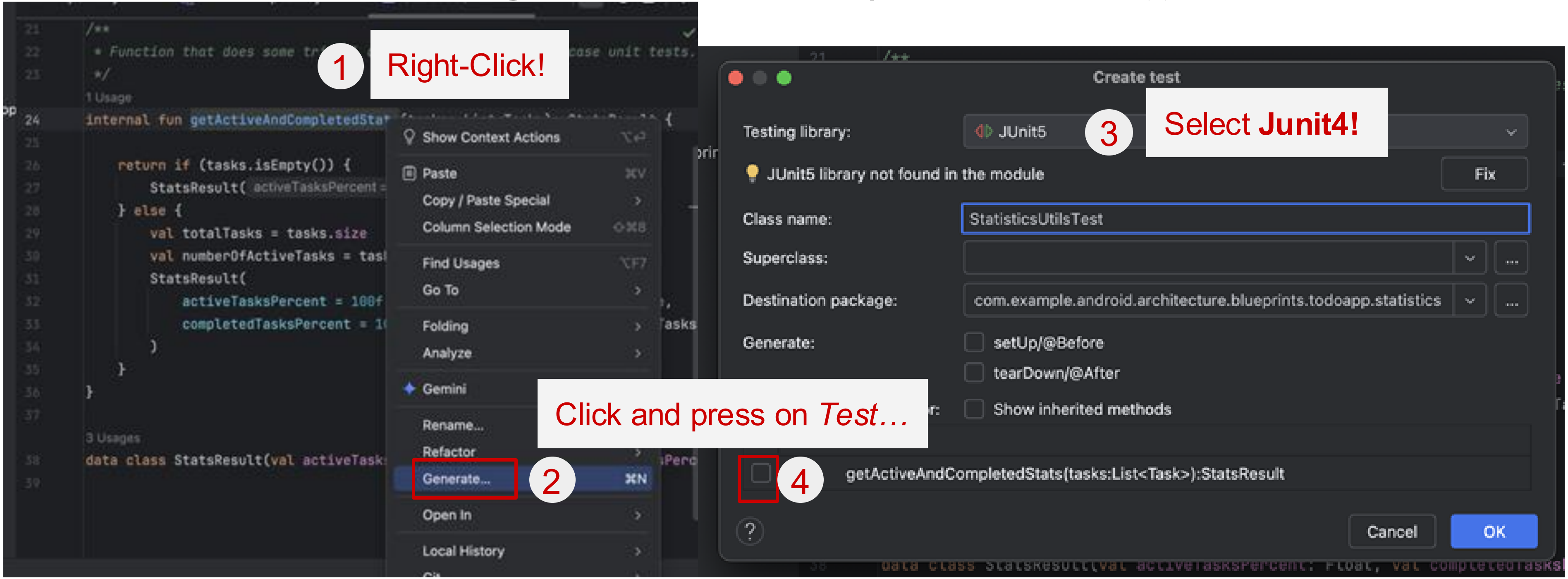
4. Generate test

other files would be generated automatically when the first test file is created



Recap: Writing Unit Tests

- We wrote a unit test for `StatisticUtils.getActiveAndCompletedStats()` last time



Recap: JUnit4

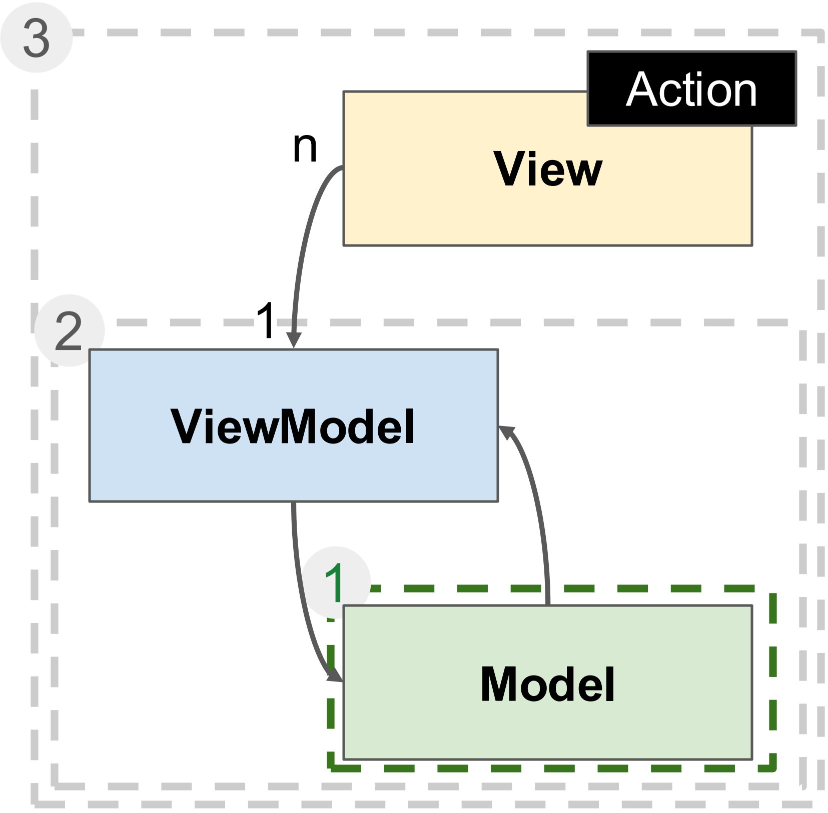
- Unit testing framework for Java and Kotlin (family of xUnit)
- Used to write and run repeatable automated unit tests
- Requires Java 6 (or higher)

JUnit annotation

JUnit assertion

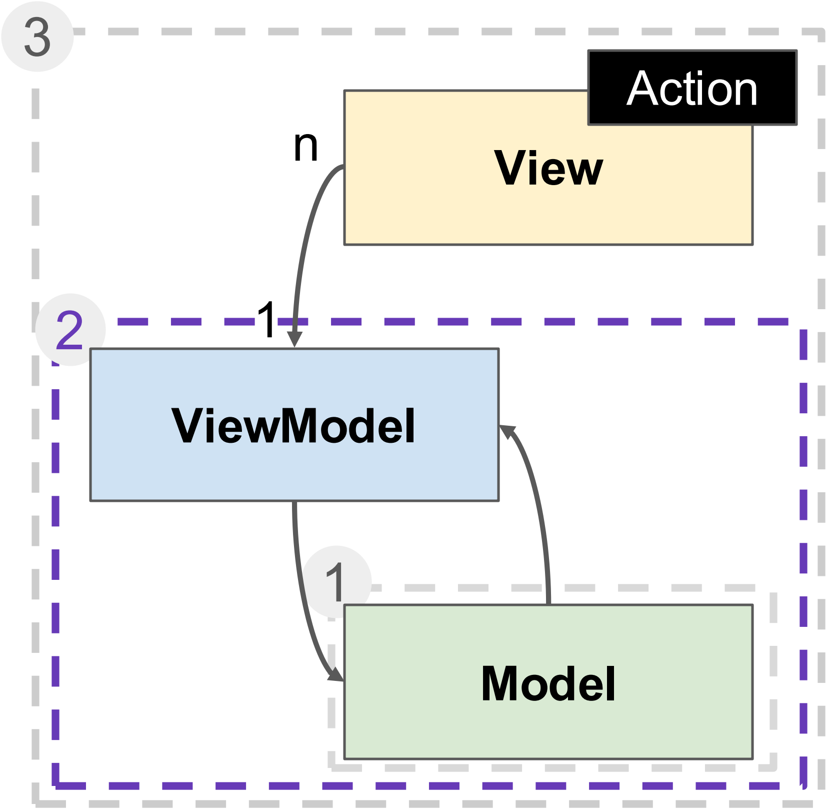
```kotlin
@Test
fun isPrime_two_returnsTrue() {
    // GIVEN input 2
    val input = 2

    // WHEN isPrime is called
    val result = isPrime(input)

    // THEN return true
    assertTrue(result)
}
```

Now We Know How to Test a Model!



Mockito Basics

How to Test a ViewModel



: Lots of data involved…

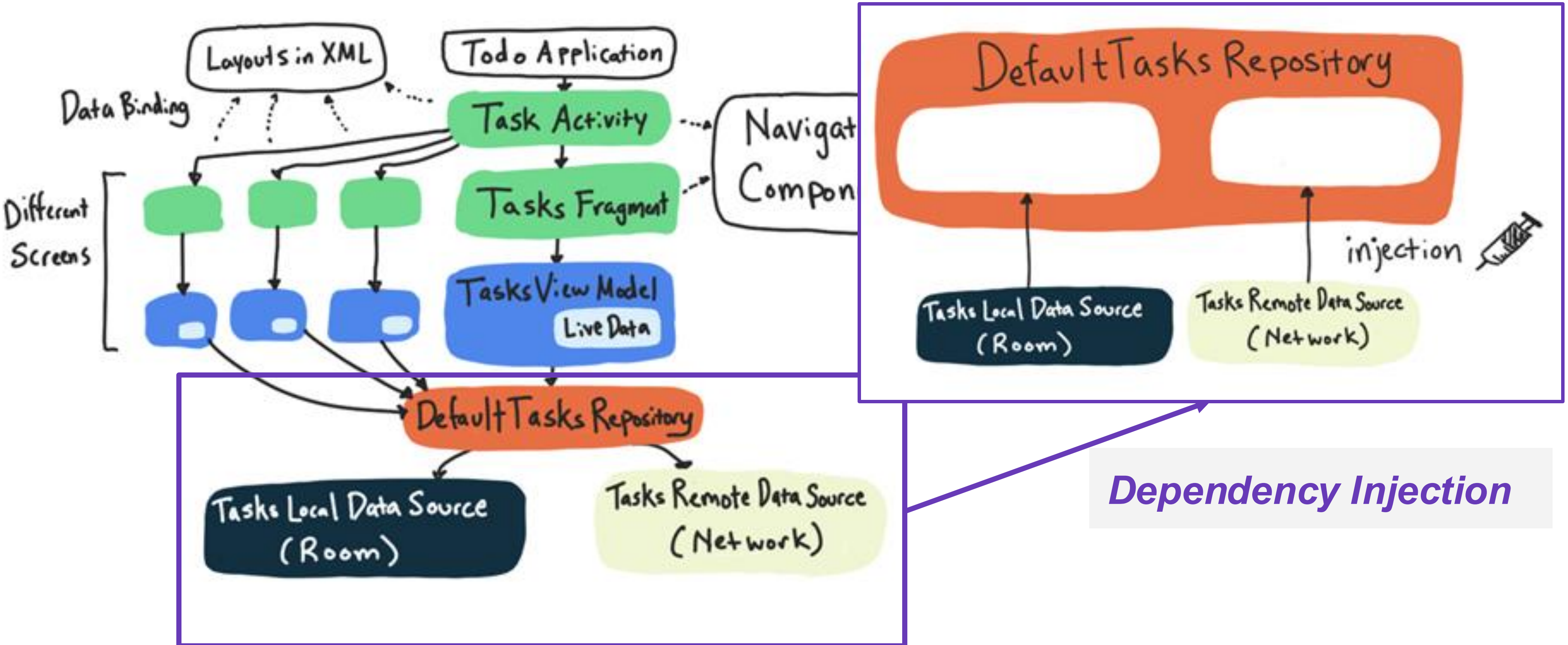
Challenges in Testing a ViewModel

```kotlin
override suspend fun createTask(title: String, description: String): String {
    // ID creation might be a complex operation so it's executed using the supplied
    // coroutine dispatcher
    val taskId = withContext( context = dispatcher) {
        UUID.randomUUID().toString()
    }
    val task = Task(
        title = title,
        description = description,
        id = taskId,
    )
    localDataSource.upsert( task = task.toLocal())
    saveTasksToNetwork()
    return taskId
}
```

Hard to implement everything in tests :(

Sol: Testing in an Isolated Environment

- Essential when testing ViewModel



Mockito

- Mocking framework for Java (Kotlin)
- Allows convenient creation of **substitutes of real objects** for testing purposes
- Used for test doubles in unit testing
- We’ll use Mockito **5.11.0**

Mockito Setup

- Add mockito dependency to your `build.gradle (app)`

  `testImplementation "org.mockito:mockito-core:5.11.0"`

  & Sync

Dependencies applied to ALL source sets

Dependencies applied to the test source set

Dependencies applied to the androidTest source set

```
dependencies {
    def lifecycle_version String = "2.6.2"
    implementation "androidx.lifecycle:lifecycle-viewmodel:$lifecycle_version"
    implementation "androidx.lifecycle:lifecycle-livedata:$lifecycle_version"
    implementation 'androidx.appcompat:appcompat:1.6.1'
    implementation 'com.google.android.material:material:1.9.0'
    implementation 'androidx.constraintlayout:constraintlayout:2.1.4'

    testImplementation 'junit:junit:4.13.2'
    testImplementation "org.mockito:mockito-core: 5.11.0

    androidTestImplementation 'androidx.test.ext:junit:1.1.5'
    androidTestImplementation "org.mockito:mockito-core:5.5.0"
    androidTestImplementation 'androidx.test.espresso:espresso-core:3.5.1'
```

Mockito Setup

- Add mockito dependency to your `build.gradle (app)`

  `testImplementation "org.mockito:mockito-core:5.11.0"`

  & Sync

```
testImplementation(libs.kotlinx.coroutines.test)
testImplementation(libs.androidx.navigation.testing)
testImplementation(libs.androidx.test.espresso.core)
testImplementation(libs.androidx.test.espresso.contrib)
testImplementation(libs.androidx.test.espresso.intents)
testImplementation(libs.google.truth)
testImplementation(libs.androidx.compose.ui.test.junit)
testImplementation("org.mockito:mockito-core:5.11.0")

// JVM tests - Hilt
testImplementation(libs.hilt.android.testing)
```

Unit Testing TasksViewModel

- Let's try to write unit test for `TasksViewModel.completeTask()`
- We want:
  - GIVEN a use case - 3 tasks, with one active and two completed
  - WHEN `completeTask` for the remaining active task is called
  - THEN change completion state & show Snackbar message

```kotlin
fun completeTask(task: Task, completed: Boolean) = viewModelScope.launch{
    if (completed) {
        taskRepository.completeTask( taskId = task.id)
        showSnackbarMessage("Task marked complete")
    } else {
        taskRepository.activateTask( taskId = task.id)
        showSnackbarMessage("Task marked active")
    }
}
```

```kotlin
6 Usages
private fun showSnackbarMessage(message: Int) {
    _userMessage.value = message
}
```

Unit Testing TasksViewModel: Problem 1

- What happens if `TasksViewModel.completeTask()` has a bug and doesn’t show the expected messages? What if `showSnackbarMessage()` is faulty?
- We're not testing `TasksViewModel.completeTask()` in an *isolated environment*

Unit Testing TasksViewModel: Problem 2

- Requires Android framework dependency (context)

```kotlin
@Test
fun completeTask_dataAndSnackbarUpdated() = runTest {
    // With a repository that has an active task
    tasksRepository = NEED_NEW_CONTEXT
    tasksViewModel = NEED_NEW_CONTEXT

    val task = Task(id = "id", title = "Title", description = "Description")
    tasksRepository.addTasks( ...tasks = task)

    // Complete task
    tasksViewModel.completeTask(task,  completed = true)

    // Verify the task is completed
    assertThat(tasksRepository.savedTasks.value[task.id]?.isCompleted).isTrue()

    // The snackbar is updated
    assertThat(tasksViewModel.uiState.first().userMessage)
        .isEqualTo("Task marked complete")
}
```

Unit Testing TasksViewModel: Solution

- **Fake:** create a class `FakeTaskRepository &` `FakeTasksViewModel` with dummy functions
  - Large cost
  - Can’t apply to Android framework classes (like `LiveData`)
- **Mock & Stub:** mock `FakeTaskRepository &` `FakeTasksViewModel` using *Mockito*!

Unit Testing TasksViewModel: Fake (1/4)

```
class TasksViewModelTest {

    // Subject under test
    32 Usages
    private lateinit var tasksViewModel: TasksViewModel

    // Use a fake repository to be injected into the viewmodel
    5 Usages
    private lateinit var tasksRepository: FakeTaskRepository

    @Before
    fun setupViewModel() {
        // We initialise the tasks to 3, with one active and two completed
        val tasksRepository = FakeTaskRepository()
        val task1 = Task(id = "1", title = "Title1", description = "Desc1")
        val task2 = Task(id = "2", title = "Title2", description = "Desc2", isCompleted = true)
        val task3 = Task(id = "3", title = "Title3", description = "Desc3", isCompleted = true)
        tasksRepository.addTasks( ...tasks = task1, task2, task3)


        tasksViewModel = TasksViewModel( taskRepository = tasksRepository, SavedStateHandle())
    }
```

1. Create and initialize fake objects

2. Declare our system under test (`TasksViewModel`)

Unit Testing TasksViewModel: Fake (2/4)

3. Write a fake implementation for every member functions

```
@Test
fun completeTask_dataAndSnackbarUpdated() = runTest {
    // With a repository that has an active task
    val task = Task(id = "id", title = "Title", description = "Description")
    tasksRepository.addTasks( ...tasks = task)

    tasksViewModel.completeTask(task,  completed = true)

    assertThat(tasksRepository.savedTasks.value[task.id]?.isCompleted).isTrue()

    assertThat(tasksViewModel.uiState.first().userMessage)
        .isEqualTo("Task marked complete")
}
```

Unit Testing TasksViewModel: Fake (3/4)

4. Check `FakeTaskRepository.kt` and check its complexity!

Compare it with `DefaultTaskRepository.kt`

Unit Testing TasksViewModel: Fake (4/4)

Fake vs DefaultTaskRepository

- All member function implemented
- **Behavior mismatch**: It skips concurrency, dispatcher usage, and data conversions (toLocal, toNetwork), so it can't validate real synchronization or data flow
- **No concurrency realism**: Running purely in-memory and synchronously hides race conditions, cancellation, and backpressure issues that appear in production

→ Let's explore others!

Unit Testing TasksViewModel: Mock

1. Create and initialize mocks

```
@RunWith( value = MockitoJUnitRunner::class)
class TasksViewModel_Simple_Test {

    5 Usages
    @Mock lateinit var taskRepository: DefaultTaskRepository

    @Test(expected = NullPointerException::class)
    fun constructingAndCollecting() = runTest {
        val vm = TasksViewModel(taskRepository, SavedStateHandle())

        vm.uiState.first()

        vm.completeTask(Task( title = "1", description = "t", isCompleted = false, id = "id"), completed = true).join()

    }
```

annotations added

Unit Testing TasksViewModel: Mock

2. Declare our system under test (`TasksViewModel`)

```
@RunWith( value = MockitoJUnitRunner::class)
class TasksViewModel_Simple_Test {

    5 Usages
    @Mock lateinit var taskRepository: DefaultTaskRepository

    @Test(expected = NullPointerException::class)
    fun constructingAndCollecting() = runTest {
        val vm = TasksViewModel(taskRepository, SavedStateHandle())

        vm.uiState.first()

        vm.completeTask(Task( title = "1", description = "t", isCompleted = false, id = "id"), completed = true).join()

    }
```

Uses the original class instead of fake

Creating and Initializing Mocks

1. Create a mock
    - `@Mock` annotation ← we used this
    - `mock()` function
2. Initialize mocks
    - `@RunWith(MockitoJUnitRunner.class)` ← we used this
    - MockitoAnnotations.openMocks(this)

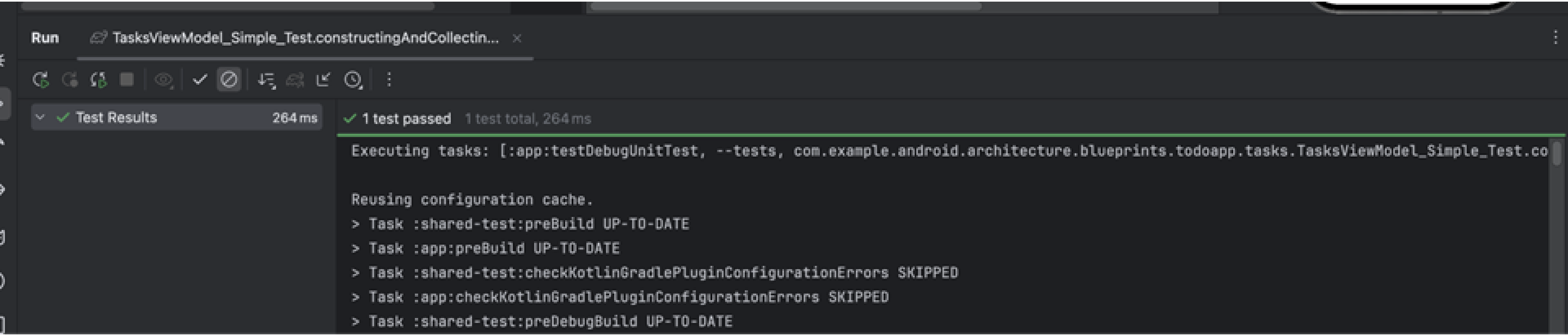
Unit Testing TasksViewModel: Mock (1/5)

- Create a class `TasksViewModel_Simple_Test`
- First test `TasksViewModel` initialization part

```
@Test
fun constructingAndCollecting()= runTest{
    // GIVEN a fresh taskRepository (already mocked: TODO)
    // WHEN a viewModel is initialized and completeTask called
    val vm = TasksViewModel(taskRepository, SavedStateHandle())
    vm.uiState.first()
    vm.completeTask(Task("1", "t", false, "id"), completed =
true).join()

    // THEN UiState is changed
    //...
}
```

Unit Testing TasksViewModel: Mock (2/5)

- Test Result: SUCCESS



Unit Testing TasksViewModel: Mock (3/5)

- Let’s test one more:

Would this work well?

```kotlin
class TasksViewModel @Inject constructor(
    private val taskRepository: TaskRepository,
    private val savedStateHandle: SavedStateHandle
) : ViewModel() {

    2 Usages
    private val _savedFilterType =
        savedStateHandle.getStateFlow( key = TASKS_FILTER_SAVED_STATE_KEY,  initialValue = ALL_TASKS)

    1 Usage
    private val _filterUiInfo = _savedFilterType.map { getFilterUiInfo( requestType = it) }.distinctUntilChanged()
    3 Usages
    private val _userMessage: MutableStateFlow<Int?> = MutableStateFlow( value = null)
    3 Usages
    private val _isLoading = MutableStateFlow( value = false)
    1 Usage
    private val _filteredTasksAsync =
        combine( flow = taskRepository.getTasksStream(),  flow2 = _savedFilterType) { tasks, type ->
            filterTasks(tasks,  filteringType = type)
        }
            .map { Async.Success( data = it) }
            .catch<Async<List<Task>>> { emit( value = Async.Error("Error while loading tasks")) }

    1 Usage
    val uiState: StateFlow<TasksUiState> = combine(
        flow = _filterUiInfo,  flow2 = _isLoading,  flow3 = _userMessage,  flow4 = _filteredTasksAsync
    ) { filterUiInfo, isLoading, userMessage, tasksAsync ->
        when (tasksAsync) {
```

Unit Testing TasksViewModel: Mock (4/5)

- Let's test one more: add a line
  - `println("FLOW TYPE >>> ${taskRepository.getTasksStream()::class}")`

```kotlin
@RunWith( value = MockitoJUnitRunner::class)
class TasksViewModel_Simple_Test {

    2 Usages
    @Mock lateinit var taskRepository: DefaultTaskRepository

    @Test
    fun constructingAndCollectingViewModel() = runTest {
        val vm = TasksViewModel(taskRepository, SavedStateHandle())
        vm.uiState.first()
        println("FLOW TYPE >>> ${taskRepository.getTasksStream()::class}")

        vm.completeTask(Task( title = "1",  description = "t",  isCompleted = false,  id = "id"), completed = true).join()

    }
```

Unit Testing TasksViewModel: Mock (5/5)

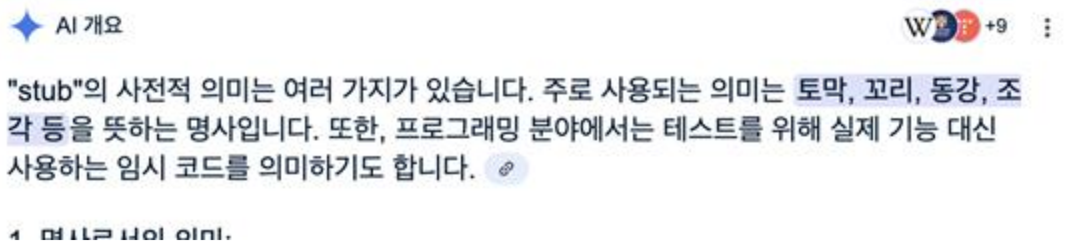
- Test Result: FAILED..

```
1 test failed   1 test total, 250 ms
WARNING: Dynamic loading of agents will be disallowed by default in a future release

Cannot invoke "Object.getClass()" because the return value of "com.example.android.architecture.blueprints.todoapp.data
 .DefaultTaskRepository.getTasksStream()" is null
java.lang.NullPointerException Create breakpoint : Cannot invoke "Object.getClass()" because the return value of "com.example.android
 .architecture.blueprints.todoapp.data.DefaultTaskRepository.getTasksStream()" is null
    at com.example.android.architecture.blueprints.todoapp.tasks.TasksViewModel_Simple_Test$constructingAndCollectingViewModel$1
     .invokeSuspend(TasksViewModelTest.kt:41)
    at com.example.android.architecture.blueprints.todoapp.tasks.TasksViewModel_Simple_Test$constructingAndCollectingViewModel$1.invoke
     (TasksViewModelTest.kt)
    at com.example.android.architecture.blueprints.todoapp.tasks.TasksViewModel_Simple_Test$constructingAndCollectingViewModel$1.invoke
     (TasksViewModelTest.kt) <1 internal line>
    at kotlin.coroutines.jvm.internal.BaseContinuationImpl.resumeWith(ContinuationImpl.kt:33) <3 internal lines>
 <61 folded frames>
```

- Why? → We need to implement some more details

Unit Testing TasksViewModel: Stub (1/4)

- `@Mock` creates a mock instance that implements the functions of `TasksRepository`
- *Unstubbed* methods return **default values (reference types ⇒ null)**
    - Other mocked functions are expected to return null as default
    - Mocked `getTasksStream()` returns `null` (!= default value “`[]`”)
- So we should ***stub*** these functions



Unit Testing TasksViewModel: Stub (2/4)

- `uiState` depends on `TaskRepository.getTasksStream()` (via `combine`).
- To test `completeTask()`, we must stub `getTasksStream()` before creating the `ViewModel` so that collecting `uiState` works.

```
@Before
fun setUp() {
    Mockito.`when`( methodCall = taskRepository.getTasksStream()).thenReturn( value = MutableStateFlow( value = emptyList()))
    vm = TasksViewModel(taskRepository, SavedStateHandle())
}

@Test
fun constructingAndCollectingViewModel() = runTest {

    vm.uiState.first()
```

Add this line

Unit Testing TasksViewModel: Stub (3/4)

```
@RunWith(MockitoJUnitRunner::class)
class TasksViewModel_With_Stub_Test {

    private lateinit var vm: TasksViewModel

    @Mock lateinit var taskRepository: DefaultTaskRepository

    @Before
    fun setUp() {
        Mockito.`when`(taskRepository.getTasksStream()).thenReturn(MutableStateFlow(emptyList()))
        vm = TasksViewModel(taskRepository, SavedStateHandle())
        println("FLOW TYPE >>> ${taskRepository.getTasksStream()::class}")
    }

    @Test
    fun constructingAndCollectingViewModel() = runTest {

        vm.uiState.first()
        vm.completeTask(Task("1", "t", false, "id"), completed = true).join()

    }
}
```

Unit Testing TasksViewModel: Stub (4/4)

- Then it works

```
ns  ✓ 1 test passed   1 test total, 274 ms
> Task :app:testDebugUnitTest
WARNING: A Java agent has been loaded dynamically (/Users/sieun/.gradle/caches/8.11.1/transforms/72ea0f62f313aa700c4c07b69bc9cd48/transfor
WARNING: If a serviceability tool is in use, please run with -XX:+EnableDynamicAgentLoading to hide this warning
WARNING: If a serviceability tool is not in use, please run with -Djdk.instrument.traceUsage for more information
WARNING: Dynamic loading of agents will be disallowed by default in a future release
FLOW TYPE >>> class kotlinx.coroutines.flow.StateFlowImpl (Kotlin reflection is not available)
class kotlinx.coroutines.flow.StateFlowImpl (Kotlin reflection is not available)
TasksViewModel_With_Stub_Test > constructingAndCollectingViewModel PASSED
BUILD SUCCESSFUL in 1s
49 actionable tasks: 1 executed, 48 up-to-date
Configuration cache entry reused.
```

Stubbing: Functions with Return Values

- `when().thenReturn()`
  - To specify a return value when called with specific params
- `when().thenThrow()`
  - To throw exceptions when function is called
- `when().thenAnswer()`
  - Allows stubbing with the generic Answer interface
  - Don't recommend using; use thenReturn() or thenThrow()

Stubbing: void Functions (1/2)

- `doThrow()`: to stub a void method with an exception
- `doAnswer()`: to stub a void method with generic Answer
- `doNothing()`: for void methods to do nothing (which is default)
- `doReturn()`: when you cannot use when(Object)
- `doCallRealMethod()`: to call the real implementation of a method

Stubbing: void Functions (2/2)

```
// Stubbing void function
doThrow(new RuntimeException()).when(mockedObject).voidFunction();
// NOT doThrow(...).when(mockedObject.voidFunction);

mockedObject.voidFunction(); // throws RuntimeException:
```

- `doSomething().when(mockedObject).function()` format:
  - Also used for:
    - stubbing methods on spy objects
    - stubbing the same method more than once, to change behavior in the middle of a test

StateFlow

- Always holds the latest value → a Hot Flow
- UI collects new values with collect { ... }
- Requires an initial value when created
- Value updates via .value (similar to LiveData.setValue()
- Compared to LiveData
  - LiveData: lifecycle-aware (Android only)
  - **StateFlow**: coroutine-based, lifecycle handling must be explicit
  - Compose-friendly - our repo also has compose-based view

StateFlow

- A simple example

```
private val _isLoading = MutableStateFlow(false)
val isLoading: StateFlow<Boolean> = _isLoading

// Update value
_isLoading.value = true

// Collect in UI
lifecycleScope.launch {
    viewModel.isLoading.collect { loading ->
        showLoading(loading)
    }
}
```

Testing StateFlow

- Add dependency: turbine (`buile.gradle.kts (:app)`)

```
dependencies {
    testImplementation(libs.androidx.compose.ui.test.junit)
    testImplementation("org.mockito:mockito-core:5.11.0")
    testImplementation("app.cash.turbine:turbine:1.1.0")
```

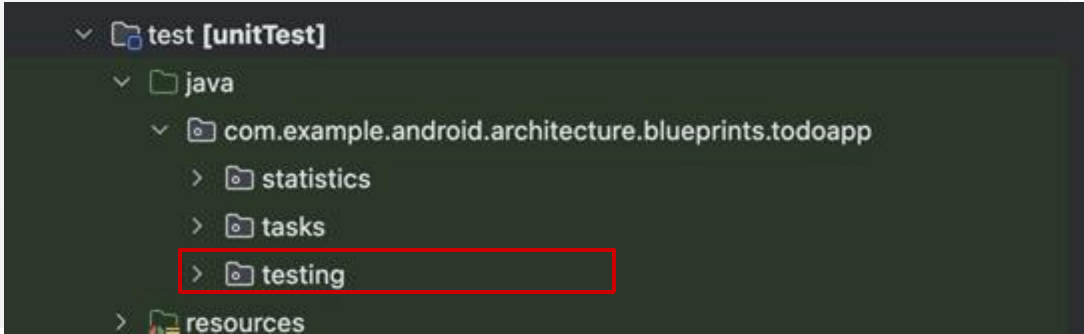
Unit Testing TasksViewModel: StateFlow (1/6)

- Let's test **`completeTask()`**, **`clearCompleteTasks()`** now
- See `TasksViewModel_StateFlow_Test()` in `TasksViewModelTest.kt` (partially implemented)
  → uncomment it

Unit Testing TasksViewModel: StateFlow (2/6)

1. To test StateFlow, we should define a dispatcher rule
   - Add testing/[MainDispatcherRule.kt]()

```kotlin
class MainDispatcherRule(
    private val dispatcher: TestDispatcher = StandardTestDispatcher()
) : TestRule {
    override fun apply(base: Statement, description: Description): Statement =
        object : Statement() {
            override fun evaluate() {
                Dispatchers.setMain(dispatcher)
                try {
                    base.evaluate()
                } finally {
                    Dispatchers.resetMain()
                }
            }
        }
}
```



Take your time fixing import errors

Unit Testing TasksViewModel: StateFlow (3/6)

- Then add rule to our test
  - `UnconfinedTestDispatcher`: for performance

```
class TasksViewModel_StateFlow_Test {

    @get:Rule val main = MainDispatcherRule(UnconfinedTestDispatcher()) // or StandardTestDispatcher()
```

Unit Testing TasksViewModel: StateFlow (4/6)

2. Stub functions (e.g., `clearCompletedTasks()`):

```
Mockito.doAnswer { inv ->
     val id = inv.arguments[0] as String
          tasksFlow.value = tasksFlow.value.filterNot { it.isCompleted }
     Unit
}.`when`(repo).clearCompletedTasks()
```

- This is needed to check if function `completeTask()`, `clearCompleteTasks()` actually react!

Unit Testing TasksViewModel: StateFlow (5/6)

3. Write a test

```
@Test
    fun clearCompletedTasks_removesCompleted_andSnackbar() = runTest {
        vm.uiState.test {
            val initial = awaitItem() // Waits and checks the value StateFlow emits

            vm.clearCompletedTasks()
            runCurrent()

            val s1 = awaitItem()
            assertEquals(R.string.completed_tasks_cleared, s1.userMessage)

            val s2 = awaitItem()
            assertTrue(s2.items.none { it.isCompleted })

            cancelAndIgnoreRemainingEvents()
        }
    }
```

Unit Testing TasksViewModel: StateFlow (6/6)

4. Check if it works

* Comment out these lines for now!

(to prevent unnecessary stub warning)

```
Mockito.`when`( methodCall = repo.getTasksStream()).thenReturn( value = tasksFlow)

//        // suspend stubs must return Unit (not null!)
//        Mockito.doAnswer { inv ->
//            val id = inv.arguments[0] as String
//            tasksFlow.value = tasksFlow.value.map { if (it.id == id) it.copy(isCompleted = false) else it }
//            Unit
//        }.`when`(repo).activateTask(Mockito.anyString())

Mockito.doAnswer {
```

3 mins

Exercise 3: Stub&Test More Functions

Stub the `completeTask()` function

: in the same way as the `activateTask()` stub in the skeleton code

Keep stub of `activateTask()` commented out when running this test!

7 mins

Exercise 4: Testing completeTask()

- Verify that marking a completed task as active updates both snackbar and list (Fill TODO in `completeTask_false_updatesSnackbar_andList()`)
- Hint
    - 1. Launch a no-op collector on vm.uiState.
    - 2. Call vm.completeTask(task, completed = false).
    - 3. Run the scheduler (runCurrent() / advanceUntilIdle()).
    - 4. Assert:
        - Snackbar message = R.string.task_marked_active.
        - Task with id "2" is active in uiState.value.items.
    - 5. Cancel the collector.

7 mins

Exercise 4: Testing `completeTask()`

IMPORTANT

- Comment out the `completeTask()` stub here
- Uncomment the `activateTask()` stub

to prevent unnecessary stub errors

Unit Testing for `TasksViewModel`: Recap

- Created **mock** for external dependency `taskRepository`
- Made `getTasksStream()` return specific values (**stub**)
- Checked that uiState's value has changed to these values (**observe StateFlow change**)

Submission

- Final code: `TasksViewModel_Final_Test` class
- Copy `TasksViewModelTest.kt` into the `submission` directory, under both `exercise03` and `exercise04` (adjusting it to each exercise's requirements)
- Make sure all tests pass successfully

Mockito: Additional Information

- Mockito `verify()`
- Argument matching
- Spy
- InjectMock

Mockito verify()

- We can use Mockito `verify()` to:
  - Check the exact number of invocations, redundant invocations
  - Check in-order calls

Mockito verify(): Number of Calls

```
//using mock
mockedList.add("once")

mockedList.add("three times")
mockedList.add("three times")
mockedList.add("three times")
```

```
//exact number of invocations verification
verify(mockedList).add("once")
verify(mockedList, times(3)).add("three times")

//verification using never(). never() is an alias to times(0)
verify(mockedList, never()).add("never happened")

//verification using atLeast()/atMost()
verify(mockedList, atMostOnce()).add("once")
verify(mockedList, atLeast(2)).add("three times")
```

Mockito verify(): In-Order Calls (1/2)

```kotlin
val firstMock = mock<List<String>>()
val secondMock = mock<List<String>>()

//using mocks
firstMock.add("was called first")
firstMock.add("was called second")
secondMock.add("was called third")
secondMock.add("was called fourth")

//create inOrder object by passing any mocks that
need to be verified in order
val inOrder = inOrder(firstMock, secondMock)
```

If order is right:

```kotlin
//verifications to assure input order
inOrder.verify(firstMock).add("was called first")
inOrder.verify(firstMock).add("was called second")
inOrder.verify(secondMock).add("was called third")
inOrder.verify(secondMock).add("was called fourth")
```

Test passes!

```
✔ Tests passed: 1 of 1 test – 1 sec 126 ms
Executing tasks: [:app:testDebugUnitTest, --tests, com.example.la
```

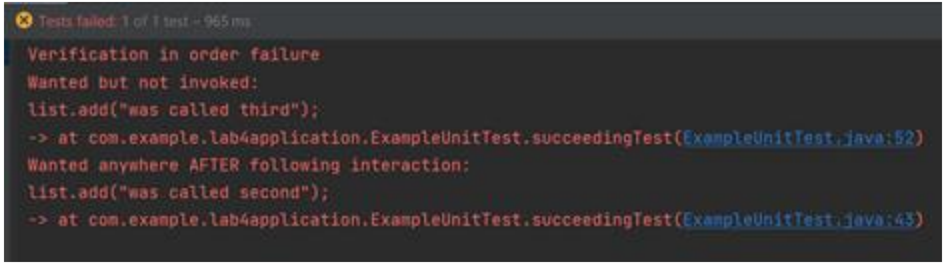
Mockito verify(): In-Order Calls (2/2)

```
List firstMock = mock(List.class);
List secondMock = mock(List.class);

//using mocks
firstMock.add("was called first");
firstMock.add("was called second");
secondMock.add("was called third");
secondMock.add("was called fourth");

//create inOrder object by passing any mocks that
need to be verified in order
val inOrder = inOrder(firstMock, secondMock);
```

If order is not right:

```
//using mocks
secondMock.add("was called fourth");
secondMock.add("was called third");
firstMock.add("was called first");

firstMock.add("was called second");
```

Error!



Argument Matchers

- Allows flexible stubbing and verification
- Common ArgumentMatchers:
  - Exact Value Matchers: `eq(value), any(), isNull()...`
  - Generic Matchers: `any()`
  - Numeric Matchers: `any<Int>(), any<Long>(), any<Double>()...`
  - Collection Matchers: `any<List<*>>(), any<Set<*>>(), any<Map<*>>()...`
-

```
whenever(mockedList.get(any())).thenReturn("element")

println(mockedList[999])

verify(mockedList).get(any())
```

Stubbing with ArgumentMatcher

Verifying with ArgumentMatcher

Source: https://javadoc.io/static/org.mockito/mockito-core/5.5.0/org/mockito/ArgumentMatchers.html

Spying with Mockito

- Spying allows you to use real objects and selectively stub certain methods
- Use the **`@Spy`** annotation or **`spy()`** method
- Recommended to use **`doReturn/Answer/Throw()`** for stubs (sometimes `when().thenReturn()` won't work)

Spying with Mockito

- Mockito does NOT delegate calls to the passed real instance, instead it creates a *copy* of it.
  - If you keep the real instance and interact with it, the spied object won't be affected
  - If an un-stubbed method is called on the *spy* but not on the real instance, there is NO effect on the real instance

Spying with Mockito: Example

- Use real method `ArrayList.add()`
- Use stubbed method `ArrayList.size()`

```kotlin
@Test
fun failingTest() {
    val list = LinkedList<String>()
    val spy = spy(list)

    // Real method is called
    // IndexOutOfBoundsException is thrown (bc list is empty)
    spy.get(0)
    assertEquals("foo", spy.get(0))
}
```

```
Index: 0, Size: 0
java.lang.IndexOutOfBoundsException : Index: 0, Size: 0
    at java.base/java.util.LinkedList.checkElementIndex(LinkedList.java:559)
    at java.base/java.util.LinkedList.get(LinkedList.java:480)
    at com.example.lab4application.ExampleUnitTest.failingTest(ExampleUnitTest.java:69) <37 internal lines>
    at jdk.proxy1/jdk.proxy1.$Proxy2.processTestClass(Unknown Source) <7 internal lines>
    at worker.org.gradle.process.internal.worker.GradleWorkerMain.run(GradleWorkerMain.java:69)
    at worker.org.gradle.process.internal.worker.GradleWorkerMain.main(GradleWorkerMain.java:74)
```

Test fails…

```kotlin
@Test
fun failingTest() {
    val list = LinkedList<String>()
    val spy = spy(list)

    // You have to use doReturn() for stubbing
    doReturn("foo").`when`(spy).get(0)
    assertEquals("foo", spy.get(0))
}
```

Test passes!

Mock vs Spy

- `@Mock` creates a fake class with empty functions. We need to stub functions else will return null
- `@Spy` creates an actual class. By default, functions maintain their functionality, and only stubbed ones are overridden
- Spies should be used carefully and occasionally
    - Real methods are invoked → unexpected results

@InjectMocks

- Automatically injects mock or spy dependencies into tested objects
- Simplifies process of injecting mocks into tested objects

```
class MyDictionary {
    Map<String, String> wordMap;

    fun MyDictionary() {
        wordMap = new HashMap<String, String>();
    }
    public void add(final String word, final String meaning) {
        wordMap.put(word, meaning);
    }
    public String getMeaning(final String word) {
        return wordMap.get(word);
    }
}
```

```
public class MyDictionaryTest {
    @Mock
    Map<String, String> wordMap;

    @InjectMocks
    MyDictionary dic = new MyDictionary();

    @Test
    public void whenUseInjectMocksAnnotation_thenCorrect() {
        Mockito.when(wordMap.get("aWord")).thenReturn("aMeaning");
        assertEquals("aMeaning", dic.getMeaning("aWord"));
    }
}
```

Automatically injected

Can stub the injected mock

Mockito Other Features

- Visit official documentation:
    - Resetting mocks
    - Using Mockito for Behavior Driven Development (BDD)

For more...

- Highly recommend reading:
  - Testing Basics codelab: full app tested, good practices in Android
  - Official Android Studio Test doc:
    - Configuration for advanced testing
    - Other types of testing (monkey test…)
  - Official Android Test doc:
    - Details & Extensions on what we've done today
    - Good practices

Hilt

- A tool to help dependency injection in Android
- Object creation and scoping, and lifecycle is automatically handled by Hilt!
- Easily integrated with Android Jetpack
- Standardized dependency injection logic
  → easy to collaborate

Hilt

```kotlin
// 의존성: Engine
class Engine @Inject constructor() {
    fun start() { println("Engine started") }
}

// Engine을 필요로 하는 Car
class Car @Inject constructor(private val engine: Engine) {
    fun drive() {
        engine.start()
        println("Car is moving!")
    }
}

// MainActivity에서 사용
@AndroidEntryPoint
class MainActivity : AppCompatActivity() {
    @Inject lateinit var car: Car

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        car.drive()
    }
}
```

When creating a Car, Hilt automatically creates and injects the required Engine.

In MainActivity, you can simply use the car variable directly.

Using Hilt

- Add dependencies

```
plugins {
  ...
  id("com.google.dagger.hilt.android") version "2.56.2" apply false
}
```

```
plugins {
  id("com.google.devtools.ksp")
  id("com.google.dagger.hilt.android")
}

android {
  ...
}

dependencies {
  implementation("com.google.dagger:hilt-android:2.56.2")
  ksp("com.google.dagger:hilt-android-compiler:2.56.2")
}
```

Using Hilt

```kotlin
// enabling Hilt
@HiltAndroidApp
class MyApplication : Application() { ... }
```

```kotlin
// provide instances with constructor
class AnalyticsAdapter @Inject constructor() { ... }
```

```kotlin
// enable Hilt in the activity
@AndroidEntryPoint
class MainActivity : AppCompatActivity() {
  @Inject lateinit var analytics: AnalyticsAdapter
  override fun onCreate(savedInstanceState: Bundle?) {
    super.onCreate(savedInstanceState)
    // analytics instance has been populated by Hilt
    // and it's ready to be used
  }
}
```

https://medium.com/androiddevelopers/dependency-injection-on-android-with-hilt-67b6031e62d

Hilt

- See this doc for details:
  - https://developer.android.com/training/dependency-injection/hilt-android
  - https://medium.com/androiddevelopers/dependency-injection-on-android-with-hilt-67b6031e62d
- Hilt is mostly implemented in the provided source code!

```
19 Usages
@HiltViewModel
class AddEditTaskViewModel @Inject constructor(
    private val taskRepository: TaskRepository,
    savedStateHandle: SavedStateHandle
) : ViewModel() {

    5 Usages
```

Summary

- Next week: Integration tests!
  - UI testing with Espresso
  - Espresso basics
  - Test coverage reports
  - Good practices when writing tests

Submission

- Zip `submission` folder and submit to eTL
- Due: **~10/23 Thu 23:59**

Thank You.
Any Questions?